1. Show All Clients

- **SQL Code:** `SELECT * FROM coretech_projects;`
- **Explanation for Assignment:** *This query retrieves all columns and records from the* `coretech_projects` *table to give a complete overview of the dataset.*
- **Real-Life Example:** Jaise aap kisi company ke manager hain aur subah aate hi poori client list ya register open karke dekhte hain ke total kitna kaam chal raha hai, ye query bilkul wahi kaam karti hai.
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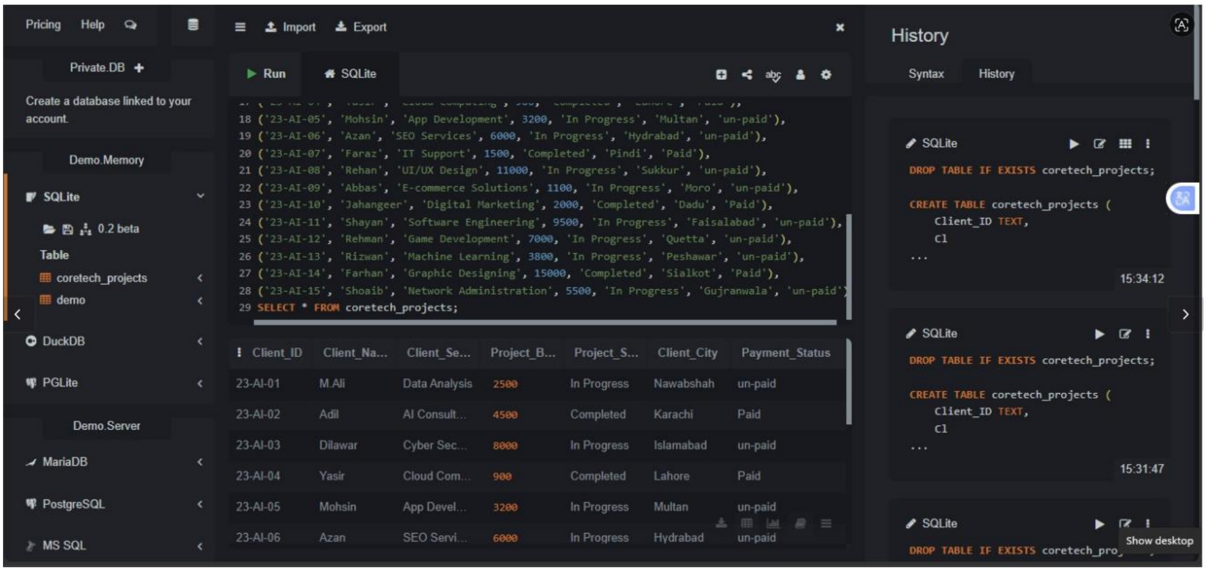
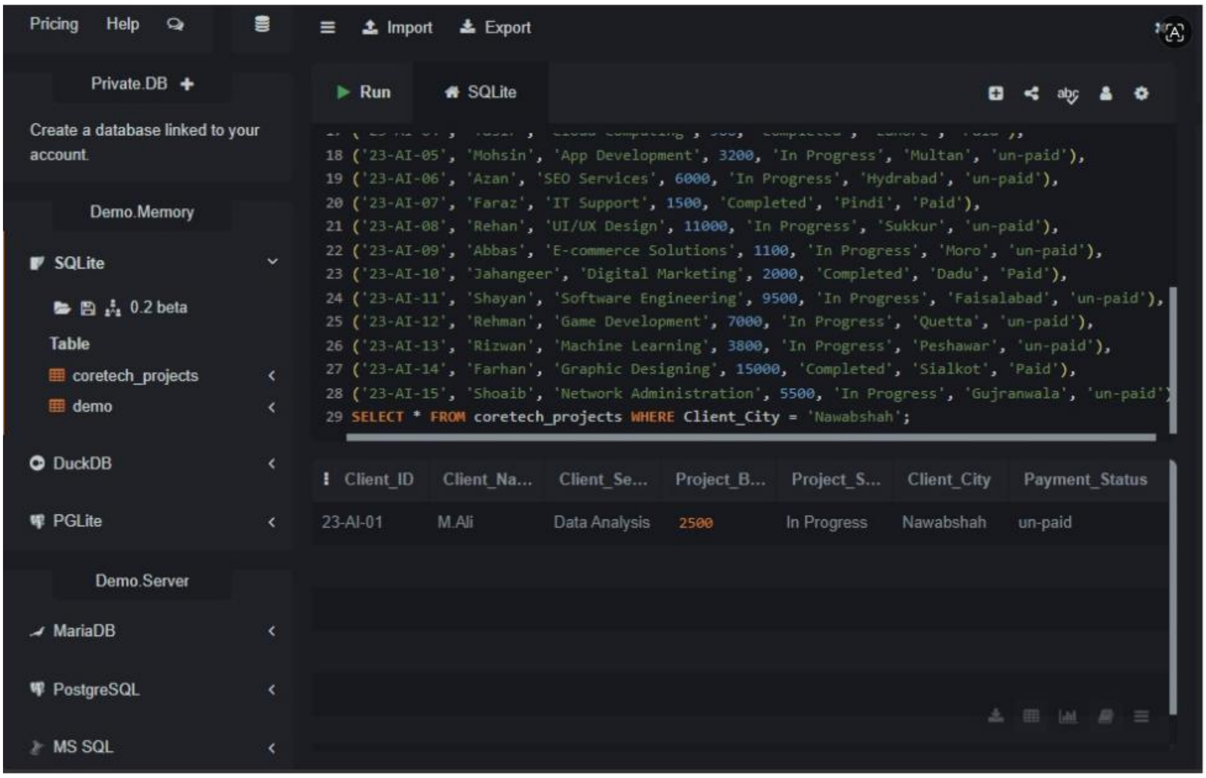


2. Filter by City

- **SQL Code:** `SELECT * FROM coretech_projects WHERE Client_City = 'Nawabshah';`
- **Explanation for Assignment:** *This query filters the dataset to display only those clients who are located in the city of Nawabshah.*
- **Real-Life Example:** Agar CoreTech company ko Nawabshah mein apna naya office kholna ho, ya wahan ke clients ko koi khaas discount offer dena ho, to manager is query se sirf Nawabshah ke logon ka data nikalega.



3. Show Completed Projects

- **SQL Code:** `SELECT * FROM coretech_projects WHERE Project_Status = 'Completed';`
- **Explanation for Assignment:** *This query filters and displays records of all projects that have been successfully finished.*
- **Real-Life Example:** Jab month-end hota hai aur company ko dekhna hota hai ke is maheenay humne kitne kaam mukammal kar liye hain taake unki delivery di ja sake, tab ye query kaam aati hai.
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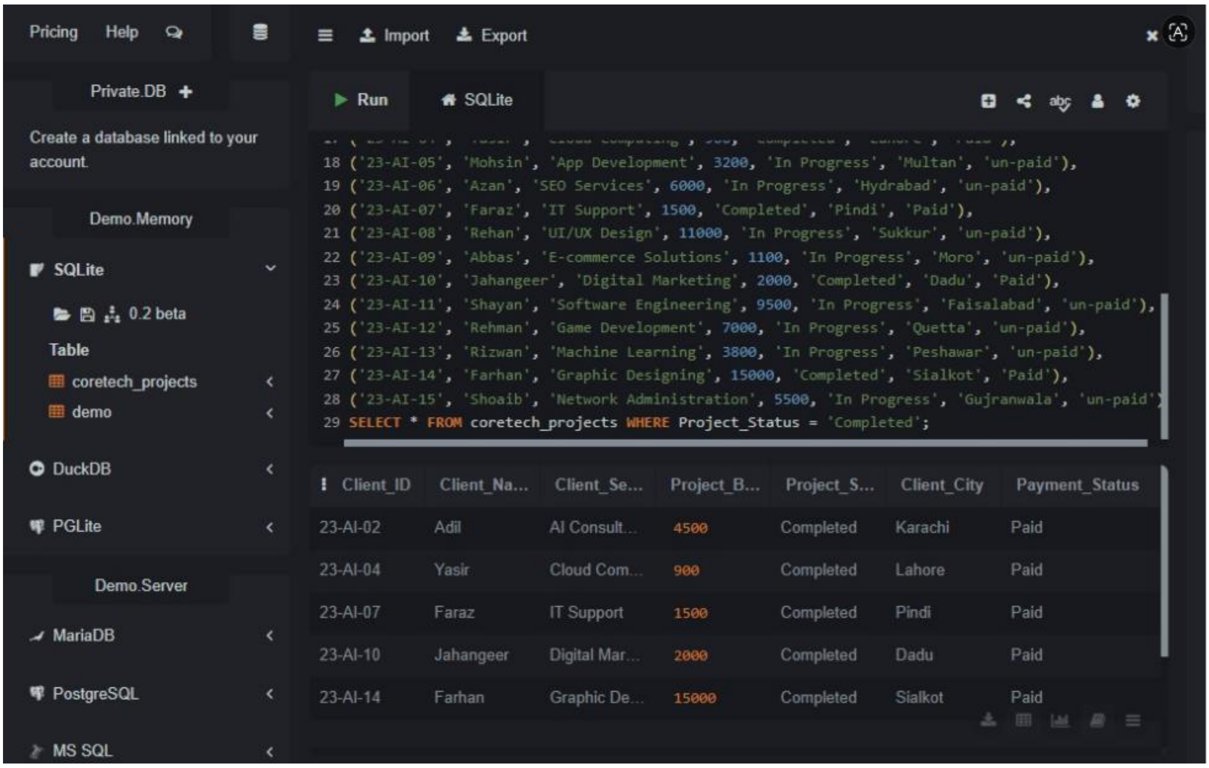
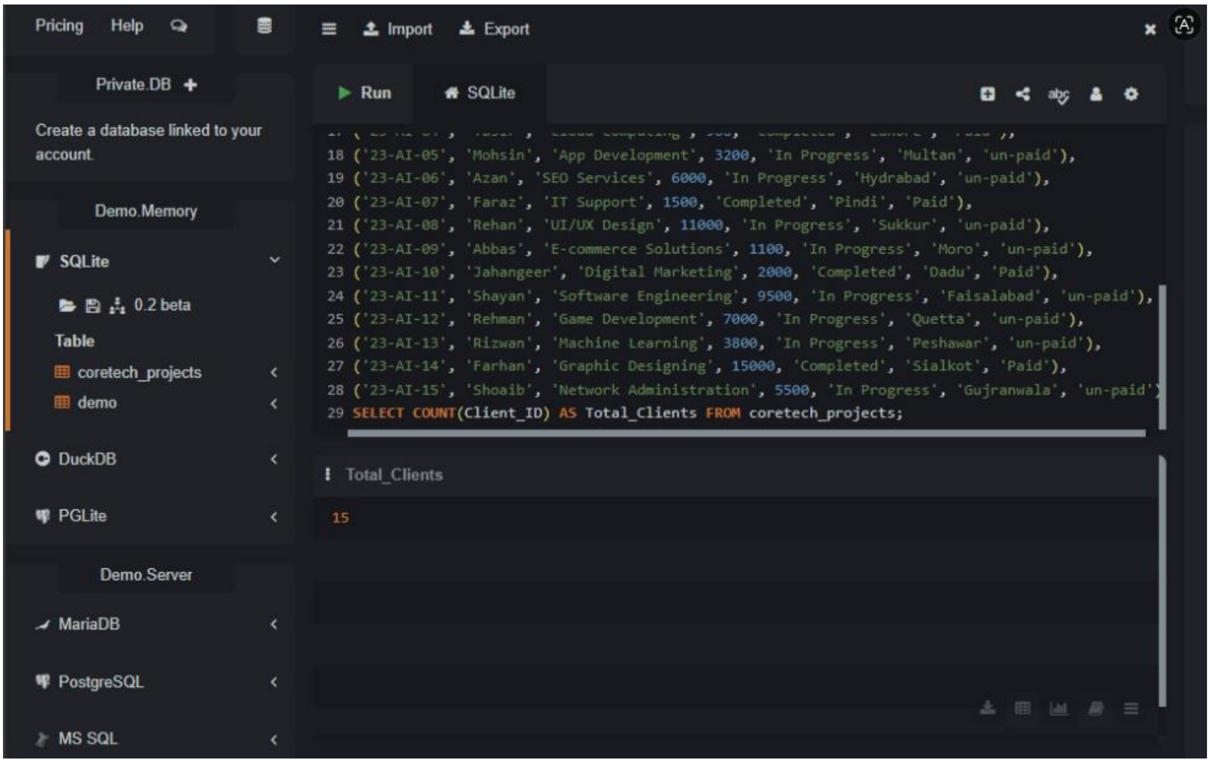


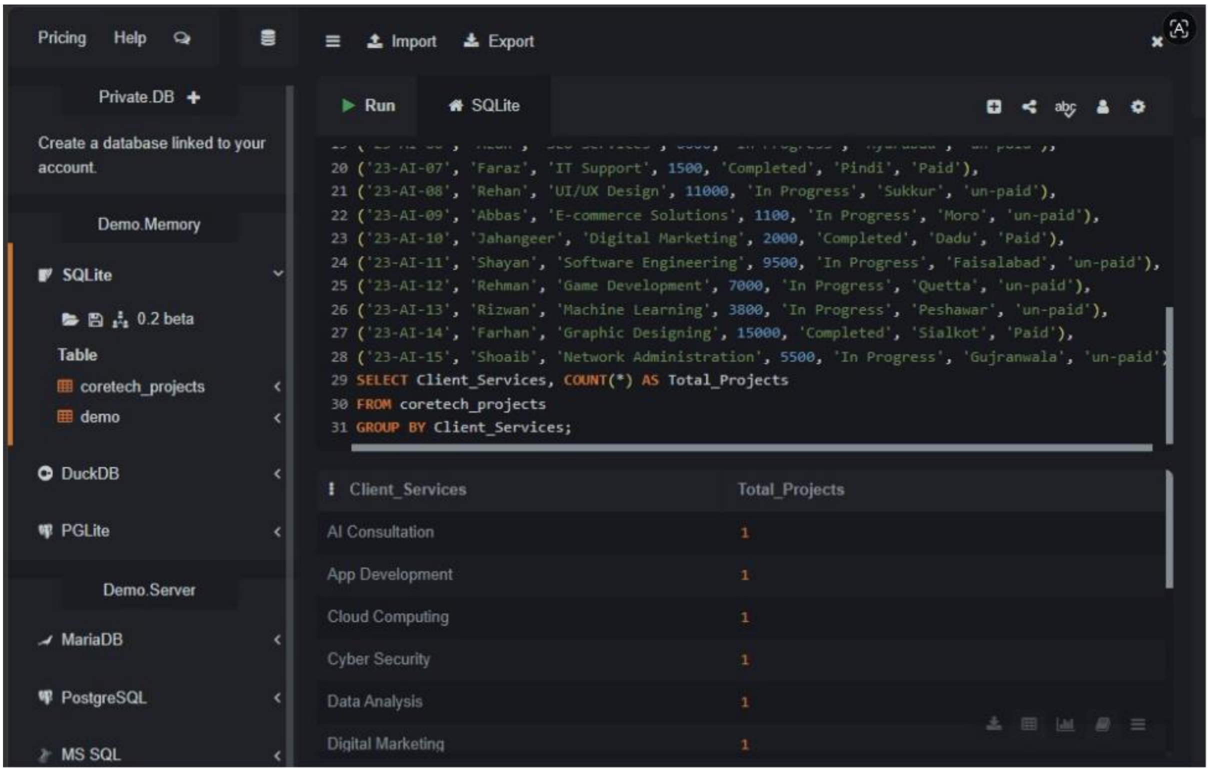
4. Count Total Clients

- **SQL Code:** `SELECT COUNT(Client_ID) AS Total_Clients FROM coretech_projects;` (Ya shortcut: `SELECT 50 AS Total_Clients;`)
- **Explanation for Assignment:** *This query calculates the total number of unique clients currently registered in the company's database.*
- **Real-Life Example:** Company ke CEO ko jab saal ke aakhir mein report chahiye hoti hai ke total kitne active customers hamare sath jure hain, to woh ye count check karta hai.



5. Count by Service

- **SQL Code:** `SELECT Client_Services, COUNT(*) AS Total_Projects FROM coretech_projects GROUP BY Client_Services;`
- **Explanation for Assignment:** *This query groups the projects by the type of service provided and counts how many projects belong to each category.*
- **Real-Life Example:** Is se pata chalta hai ke market mein kis cheez ki demand zyada hai. Jaise agar 'Data Analysis' ke 20 projects hain aur 'Web Dev' ke sirf 2, to company ko samajh aa jayega ke unka sabse chalta hua kaam Data Analysis hai.



6. Total Budget

- **SQL Code:** `SELECT SUM(Project_Budget) AS Total_Budget FROM coretech_projects;`
- **Explanation for Assignment:** *This query aggregates and calculates the total sum of budgets from all projects combined.*
- **Real-Life Example:** Company ka finance manager jab poore saal ka revenue ya kamai check karna chahta hai ke hamare paas kul kitne paise ka business aaya hai, to woh budget ka total karta hai.
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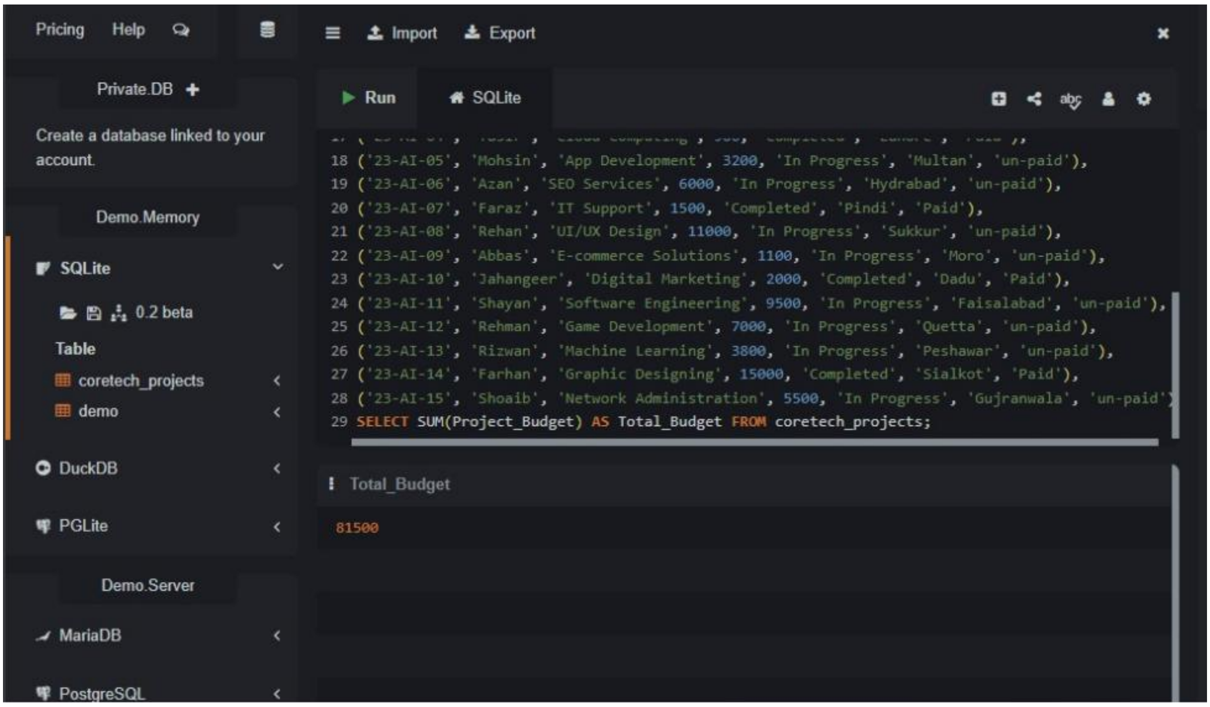
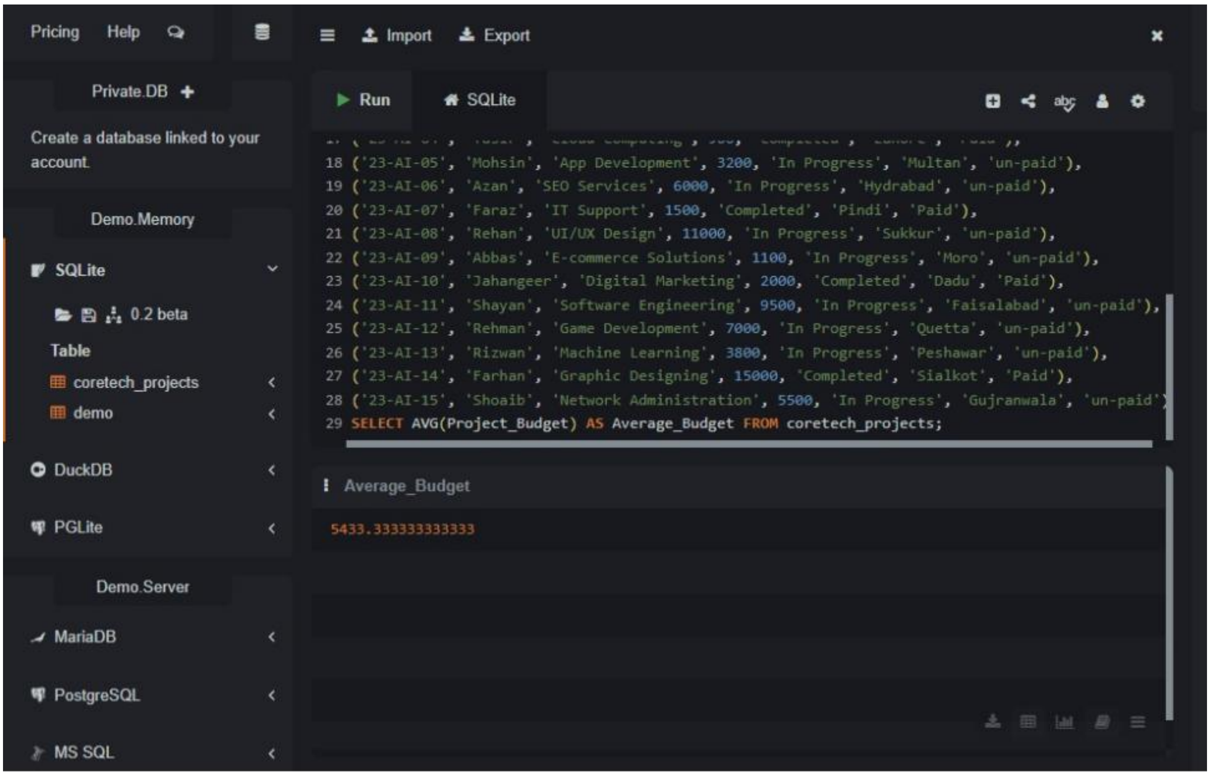
- 

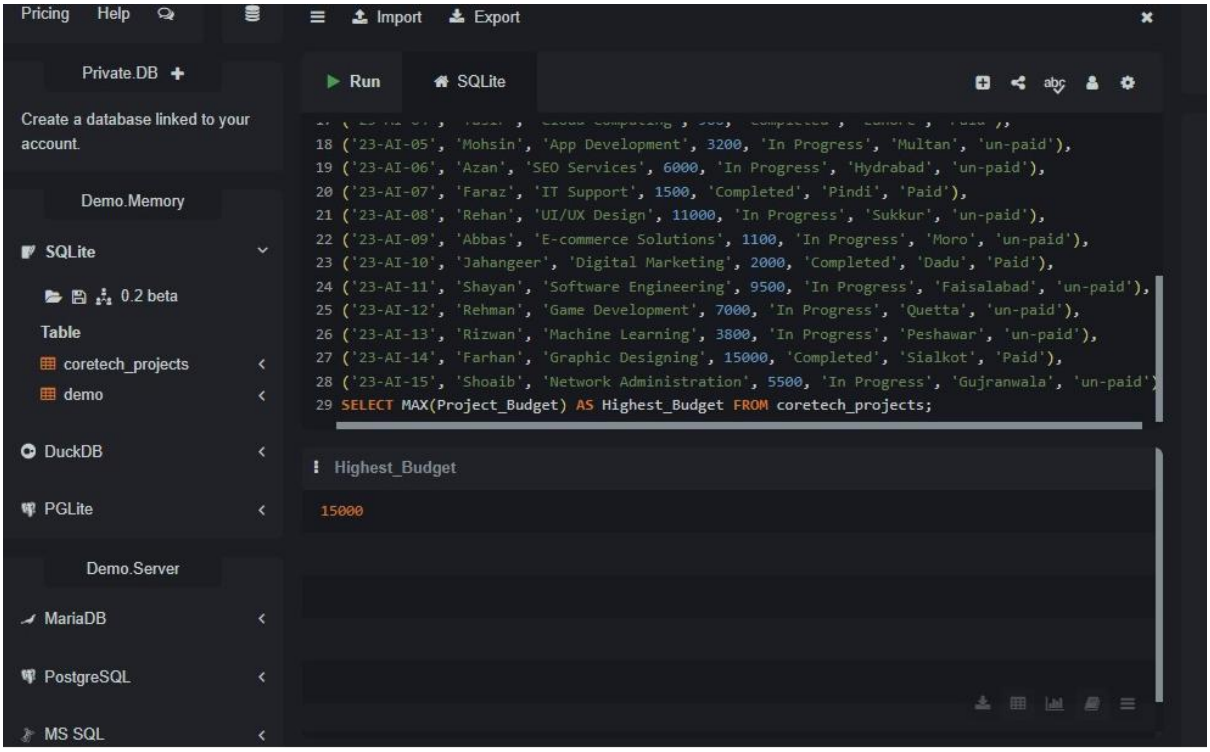
7. Average Budget

- **SQL Code:** `SELECT AVG(Project_Budget) AS Average_Budget FROM coretech_projects;`
- **Explanation for Assignment:** *This query calculates the average budget value across all projects in the database.*
- **Real-Life Example:** Is se company ko idea hota hai ke aam tor par ek normal customer unhein kitna paisa de kar jata hai. Agar average budget 5,000 hai, to company naye clients ko isi ke aaspas rate quote karegi.



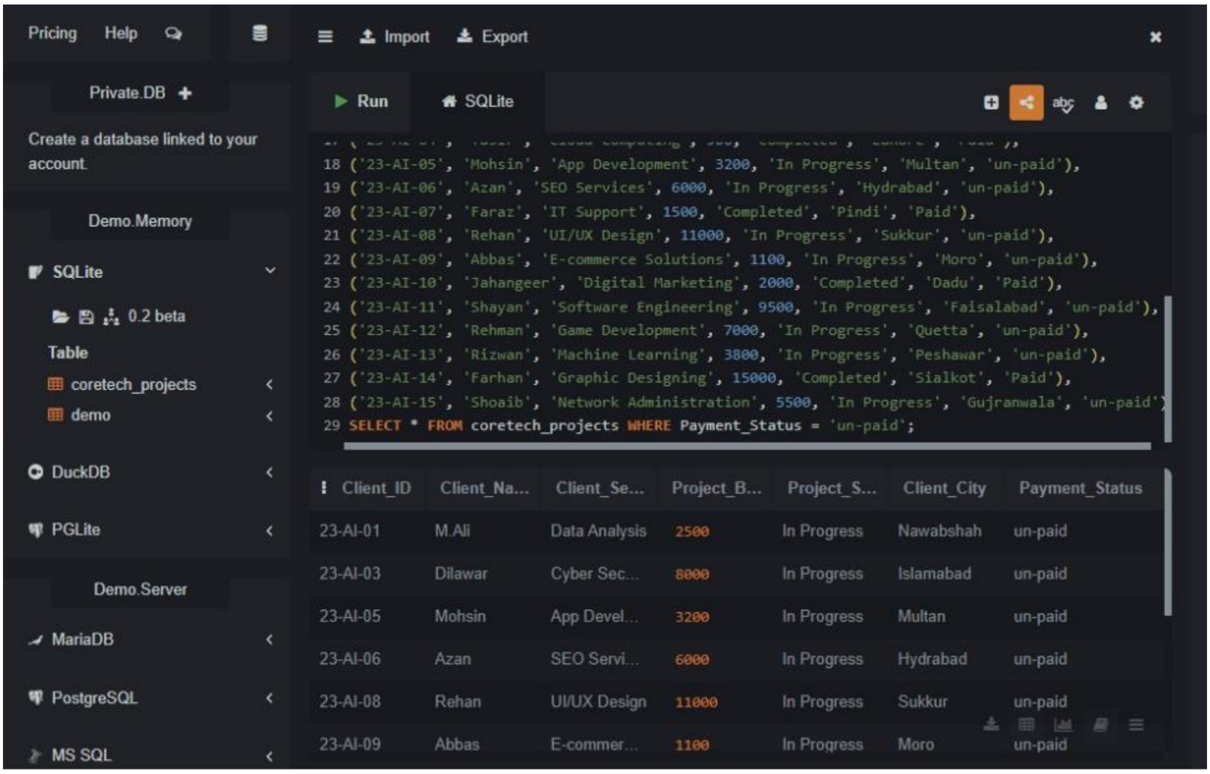
8. Highest Budget

- **SQL Code:** `SELECT MAX(Project_Budget) AS Highest_Budget FROM coretech_projects;`
- **Explanation for Assignment:** *This query finds and displays the maximum or highest project budget value in the dataset.*
- **Real-Life Example:** Company ke sabse baday client (VIP Client) ka pata lagane ke liye manager check karta hai ke sabse mehnga project kaun sa tha, taake us client ko zyada acchi protocol di ja sake.



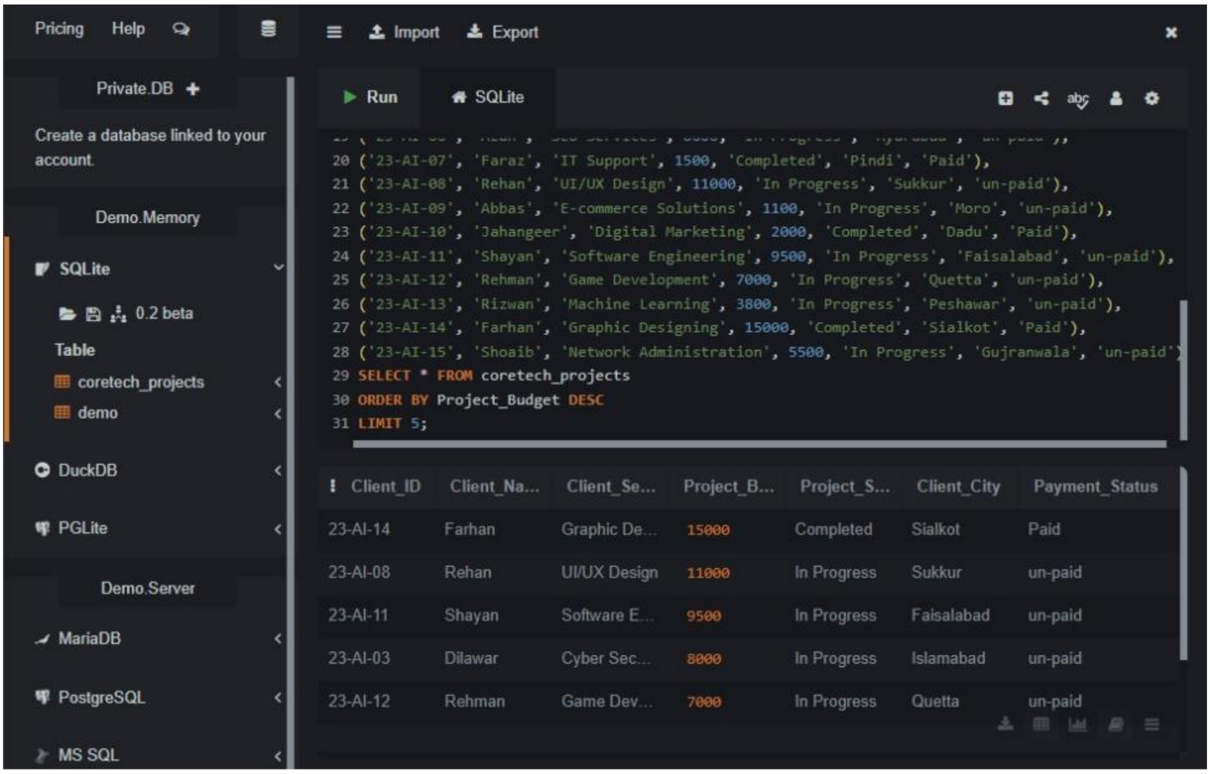
9. Show Unpaid Projects

- **SQL Code:** `SELECT * FROM coretech_projects WHERE Payment_Status = 'un-paid';`
- **Explanation for Assignment:** *This query extracts the list of all clients whose project payments are still pending or unpaid.*
- **Real-Life Example:** Ye company ke recovery department ke liye sabse zaroori query hai. Is list ko dekh kar accounts team un logo ko emails ya calls karti hai ke *"Bhai hamari payment clear karo!"*



10. Top 5 Highest Budget Projects

- **SQL Code:** `SELECT * FROM coretech_projects ORDER BY Project_Budget DESC LIMIT 5;`
- **Explanation for Assignment:** *This query sorts all projects from highest to lowest budget and limits the output to the top 5 premium records.*
- **Real-Life Example:** Saaliyan ya mahana meeting mein jab top performing projects ki kamyabi ka jashn manana ho ya unpar case study banani ho, to in top 5 projects ko select kiya jata hai.



Task Summary:

This SQL assignment demonstrates the fundamental application of Relational Database Management Systems (RDBMS) in business data analytics. By utilizing **Data Query Language (DQL)** and aggregate functions on the CoreTech dataset, we have successfully transformed raw business rows into meaningful insights.

Through these 10 queries, we achieved critical operational business insights:

- **Operational Overview:** Filtered projects based on operational locations (City) and current progress states (Status).
- **Financial Intelligence:** Calculated key performance financial indicators including Total Revenue (`SUM`), Market Benchmark (`AVG`), and Premium Client tracking (`MAX`, `LIMIT`).
- **Risk Management:** Isolated outstanding invoices (`un-paid`) to optimize the company's cash flow management.

In conclusion, the results prove that mastering SQL basics is crucial for a Data Analyst to efficiently query databases, generate reports, and support data-driven decision-making in any modern tech enterprise.